

第十周作业

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1. 习题五:

8.

(1) 令 X_i = “第 i 位居民每日用电量”, $X_i \sim U(0, 20)$, $E(X_i) = \frac{0+20}{2} = 10$, $D(X_i) = \frac{(20-0)^2}{12} = \frac{100}{3}$

$$P(\sum_{i=1}^{7500} X_i > 76000) = 1 - P(\sum_{i=1}^{7500} X_i \leq 76000) = 1 - P\left(\frac{\sum_{i=1}^{7500} X_i - 7500 \times 10}{\sqrt{7500 \times \frac{100}{3}}} \leq \frac{76000 - 10 \times 7500}{\sqrt{7500 \times \frac{100}{3}}}\right) = 1 - \Phi(2) = 0.0228$$

(2) 设电力公司每天至少需向该地区供应 $x kW \cdot h$ 的电

$$P(\sum_{i=1}^{7500} X_i \leq x) = P\left(\frac{\sum_{i=1}^{7500} X_i - 7500 \times 10}{\sqrt{7500 \times \frac{100}{3}}} \leq \frac{x - 10 \times 7500}{\sqrt{7500 \times \frac{100}{3}}}\right) = \Phi\left(\frac{x - 10 \times 7500}{\sqrt{7500 \times \frac{100}{3}}}\right) = 0.999$$

$$\therefore \frac{x - 10 \times 7500}{\sqrt{7500 \times \frac{100}{3}}} = 3.1$$

$\therefore x = 76550$, 电力公司每天至少需向该地区供应 $76550 kW \cdot h$ 的电

9.

(1) 设 X_i = “第 i 周的销售量”, $E(X_i) = 0 \times 0.2 + 1 \times 0.6 + 2 \times 0.2 = 1$, $D(X_i) = 0.2 \times (0 - 1)^2 + 0.6 \times (1 - 1)^2 + 0.2 \times (2 - 1)^2 = 0.4$

$$E(\sum_{i=0}^{52} X_i) = \sum_{i=0}^{52} E(X_i) = 52, D(\sum_{i=0}^{52} X_i) = \sum_{i=0}^{52} D(X_i) = 20.8$$

$$P(42 \leq \sum_{i=0}^{52} X_i \leq 62) = P(|\sum_{i=0}^{52} X_i - E(\sum_{i=0}^{52} X_i)| < 10) \geq 1 - \frac{20.8}{10^2} = 0.792$$

$$(2) P(42 \leq \sum_{i=0}^{52} X_i \leq 62) = P(\sum_{i=0}^{52} X_i \leq 62) - P(\sum_{i=0}^{52} X_i \leq 42) = P\left(\frac{\sum_{i=0}^{52} X_i - 1 \times 52}{\sqrt{52 \times 0.4}} \leq \frac{62 - 52}{\sqrt{52 \times 0.4}}\right) - P\left(\frac{\sum_{i=0}^{52} X_i - 1 \times 52}{\sqrt{52 \times 0.4}} \leq \frac{42 - 52}{\sqrt{52 \times 0.4}}\right) = \Phi(2.19) - \Phi(-2.19) = 0.9714$$

11.

$$(1) \int_{-\infty}^{+\infty} f(x) dx = 2 \int_0^{30} k(30 - (30 - x)) dx = 900k = 1$$

$$\therefore k = \frac{1}{900}$$

$$(2) \text{ 设 } X_i = \text{“第 } i \text{ 位顾客的消费额”}, E(X_i) = \int_{-\infty}^{+\infty} xf(x) dx = \frac{1}{900} (\int_0^{30} x^2 dx + \int_{30}^{60} (60 - x)x dx) = 30$$

$$E(X_i^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx = \frac{1}{900} (\int_0^{30} x^3 dx + \int_{30}^{60} (60 - x)x^2 dx) = 1050, D(X_i) = E(X_i^2) - E^2(X_i) = 150$$

$$\begin{aligned}
2 \quad & \therefore E(\sum_{i=0}^{200} X_i) = \sum_{i=0}^{200} E(X_i) = 6000, D(\sum_{i=0}^{200} X_i) = \sum_{i=0}^{200} D(X_i) = 30000 \quad P(5800 \leq \sum_{i=0}^{200} X_i \leq 6200) = \\
& P(|\sum_{i=0}^{200} X_i - E(\sum_{i=0}^{200} X_i)| < 200) \geq 1 - \frac{30000}{200^2} = 0.25 \\
& (3) P(5800 \leq \sum_{i=0}^{200} X_i \leq 6200) = P(\sum_{i=0}^{200} X_i \leq 6200) - P(\sum_{i=0}^{200} X_i \leq 5800) = P(\frac{\sum_{i=0}^{200} X_i - 200 \times 30}{\sqrt{200 \times 150}} \leq \\
& \frac{6200 - 6000}{\sqrt{200 \times 150}}) - P(\frac{\sum_{i=0}^{200} X_i - 200 \times 30}{\sqrt{200 \times 150}} \leq \frac{5800 - 6000}{\sqrt{200 \times 150}}) = \Phi(1.15) - \Phi(-1.15) = 0.7498
\end{aligned}$$

2. 习题六:

2.

(1), (2), (3), (7)

5.

$$(1) P(\max\{X_1, X_2, X_3\} < 5) = P(X_1 < 5)P(X_2 < 5)P(X_3 < 5) = \Phi(1)^3 = 0.5955$$

$$(2) P(\{-2.5 < X_1 < 3.5\} \cup \{2 < X_3 < 6.5\}) = P(-2.5 < X_1 < 3.5) + P(2 < X_3 < 6.5) - P(\{-2.5 < X_1 < 3.5\} \cap \{2 < X_3 < 6.5\}) = 0.7873$$

$$(3) E(X_1^2 X_2^2 X_3^2) = E(X_1^2)E(X_2^2)E(X_3^2) = (D(X_1) + E^2(X_1))^3 = 2197$$

$$(4) D(X_1 X_2 X_3) = E(X_1^2 X_2^2 X_3^2) - E^2(X_1 X_2 X_3) = E(X_1^2 X_2^2 X_3^2) - (E(X_1)E(X_2)E(X_3))^2 = 2133$$

$$D(2X_1 - 3X_2 - X_3) = 4D(X_1) + 9D(X_2) + D(X_3) = 126$$

6.

$$\begin{aligned}
P(|\bar{X} - \mu| < 1) &= P(|\sum_{i=1}^n X_i - n\mu| < n) = P(\frac{\sum_{i=1}^n X_i - n\mu}{6\sqrt{n}} \leq \frac{n}{6\sqrt{n}}) - P(\frac{\sum_{i=1}^n X_i - n\mu}{6\sqrt{n}} \leq -\frac{n}{6\sqrt{n}}) = \Phi(\frac{\sqrt{n}}{6}) - \\
&\Phi(-\frac{\sqrt{n}}{6}) = 2\Phi(\frac{\sqrt{n}}{6}) \geq 0.95
\end{aligned}$$

$$\therefore n \geq 138.2976$$

$$\therefore n \geq 139 \text{ 时, } P(|\bar{X} - \mu| < 1) \geq 0.95$$

7.

由题意得 $X_1, X_2, \dots, X_n$ 相互独立, $X_i \sim N(\mu, \sigma^2), \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$Cov(X_i - \bar{X}, X_j - \bar{X}) = Cov(X_i, X_j) - Cov(X_i, \bar{X}) - Cov(X_j, \bar{X}) + D(\bar{X}) = Cov(X_i, X_j) - \frac{1}{n}D(X_i) -$$

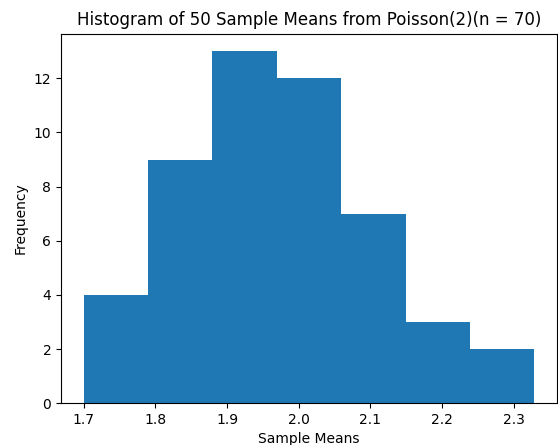
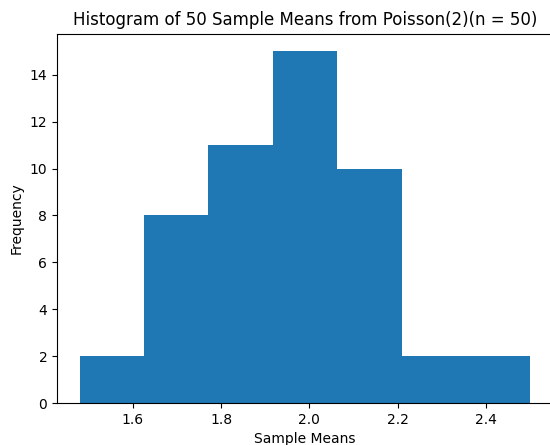
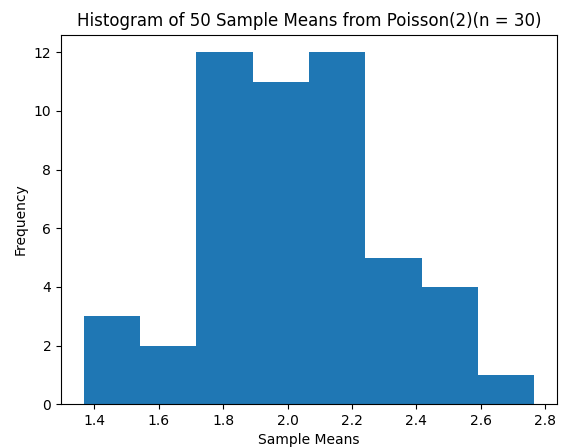
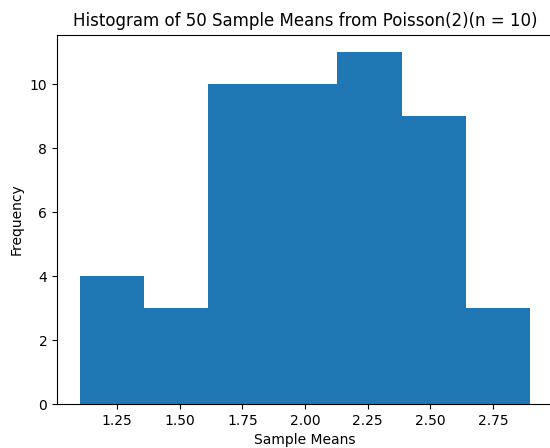
$$\frac{1}{n}D(X_j) + D(\bar{X}) = \begin{cases} \frac{n-1}{n}\sigma^2, & i = j \\ -\frac{1}{n}\sigma^2, & i \neq j \end{cases}$$

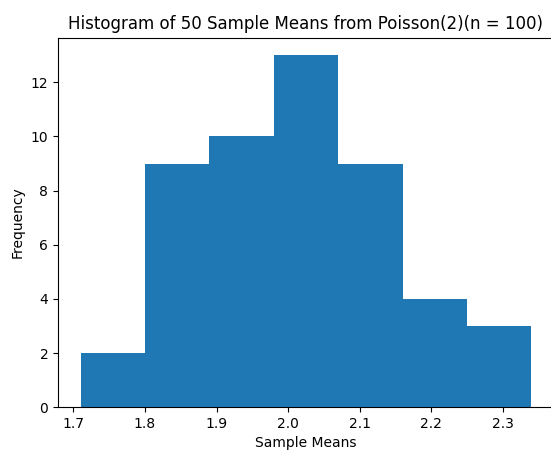
$$D(X_i - \bar{X}) = D(X_i - \frac{\sum_{i=1}^n X_i}{n}) = \frac{1}{n^2}D(-X_1 - X_2 - \dots + (n-1)X_i - X_{i+1} - \dots - X_n) = \frac{n-1}{n}\sigma^2$$

$$D(X_j - \bar{X}) = D(X_i - \bar{X})$$

$$\rho = \frac{Cov(X_i - \bar{X}, X_j - \bar{X})}{\sqrt{D(X_i - \bar{X})}\sqrt{D(X_j - \bar{X})}} = \begin{cases} 1, & i = j \\ -\frac{1}{n-1}, & i \neq j \end{cases}$$

开放式案例分析题：





(2) 直方图呈现在 2.0 附近处出现“峰”的特征

(3) 随着 n 的增大，直方图在 2.0 附近的“峰”越来越明显。这是因为随着 n 的增大，样本均值靠近到总体均值的概率越大。